

PRIYANSHU SINGH

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SUMMARY

Software Engineer with experience building AI-powered SaaS applications, scalable backend systems, and cloud-native platforms. Experienced in designing multi-agent AI workflows using OpenAI APIs and LangGraph while developing production-ready applications with Python, FastAPI, React, Next.js, PostgreSQL, Docker, Kubernetes, and AWS. Passionate about distributed systems, intelligent automation, and building highly scalable software through clean engineering practices.

EDUCATION

Vellore Institute of Technology, Bhopal

Sep 2023 – May 2027

B.Tech in Computer Science and Engineering

CGPA: 8.89/10

WORK EXPERIENCE

TM Cloud Solutions LLC (USA)

Jun 2026 – Present

AI Application Developer Intern

Remote

- Engineered Resume2Hire, an AI-powered SaaS platform that generates ATS-optimized resumes by orchestrating 2-3 LLM calls through a multi-agent LangGraph pipeline, significantly reducing redundant API usage while improving generation efficiency.
- Designed and implemented a modular backend using FastAPI, asynchronous background jobs, and PostgreSQL, enabling scalable resume parsing, job processing, and document generation workflows.
- Built an end-to-end AI orchestration pipeline integrating OpenAI GPT-4o, LangGraph, ATS evaluation, gap analysis, resume optimization, and automated Latex rendering into PDF, DOCX, and TEX formats.
- Developed responsive frontend modules using React, Tailwind CSS, and Framer Motion, delivering an interactive user experience with real-time workflow tracking, template management, and AI-assisted resume editing.
- Implemented browser automation using Playwright to automate multi-step job application workflows, reducing manual effort and improving the reliability of repetitive recruitment processes.
- Collaborated on system architecture, prompt engineering, AI agent design, testing, debugging, and production optimization while following Agile development practices and secure software engineering standards.

PROJECTS

Virtual Pandit

Production Deployed

AI-Powered Spiritual Guidance Platform (Next.js, TypeScript, MongoDB, LangChain, Groq)

- Built an AI-powered spiritual guidance platform leveraging Large Language Models to deliver contextual responses from Vedic knowledge sources.
- Implemented RAG pipelines using LangChain and vector embeddings, improving contextual response relevance by over 40%.
- Built semantic search across 1000+ scripture passages using vector embeddings and LangChain-powered retrieval.
- Integrated external APIs for astrology, Panchang generation, and scripture services to provide real-time personalized experiences.
- Designed serverless backend APIs using Next.js and deployed production workloads on AWS cloud infrastructure.
- Optimized performance using Server-Side Rendering and Incremental Static Regeneration to improve SEO and page responsiveness.

WinkWear

Production Deployed

AI-Enabled MERN E-commerce Platform (React, Node.js, MongoDB, Gemini, AWS)

- Deployed and maintained a production-grade e-commerce platform on AWS, continuously serving users for over 6 months.
- Configured Google Search Console, sitemap indexing, and SEO optimization to improve search visibility and discoverability.
- Achieved search engine indexing with 150+ organic impressions and maintained a top-5 average search ranking position.
- Designed and secured 10+ REST APIs using JWT authentication and role-based authorization mechanisms.
- Integrated Gemini AI to generate personalized product recommendations with average response times below 2 seconds.
- Implemented contextual product retrieval and persistent cart management to enhance user shopping experience.

Mavelo

Production Deployed

Cloud-Native Vehicle Rental Platform (React, Node.js, AWS, Docker, Kubernetes)

- Developed a full-stack vehicle rental platform supporting dynamic search, booking workflows, inventory management, and real-time availability tracking.
- Containerized frontend and backend services using Docker and orchestrated deployments on Kubernetes for scalable and fault-tolerant operations.
- Leveraged AWS services including EC2, S3, CloudFront, IAM, and Route53 to build a secure and highly available cloud infrastructure.
- Implemented CI/CD pipelines using GitHub Actions, reducing release cycles from hours to minutes.
- Configured Prometheus and Grafana dashboards for real-time monitoring, alerting, resource tracking, and application observability.
- Enhanced platform security through JWT authentication, rate limiting, secure headers, and role-based access control mechanisms.

BlogVerse

Production Deployed

Cloud-Native Publishing SaaS Platform (AWS EKS, Docker, Kubernetes, GitHub Actions)

- Architected a highly available cloud-native publishing platform using a three-tier architecture with React, Express.js, MongoDB, and scalable service separation.
- Containerized and deployed 10+ microservices on Amazon EKS using Kubernetes, enabling scalable and highly available workloads.
- Automated build, security scanning, and deployment workflows using GitHub Actions, reducing manual deployment effort by 80%.
- Provisioned and configured AWS infrastructure including EKS, ECR, IAM, Application Load Balancers, and networking components.
- Integrated Prometheus, Grafana to monitor 20+ infrastructure metrics and centralized application logs.
- Applied enterprise-grade security practices including JWT authentication, WAF integration, and secure containerized deployments.

SKILLS

Programming: C++, Python, JavaScript, TypeScript, SQL, Shell Scripting

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Framer Motion

Backend: FastAPI, Node.js, Express.js, REST APIs, Microservices

AI: OpenAI API, LangGraph, AI Agents, Prompt Engineering, Playwright

Cloud: AWS, Microsoft Azure

DevOps: Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Ansible

Databases: PostgreSQL, MongoDB, MySQL

Tools: Git, GitHub, Linux, VS Code, Postman

ACHIEVEMENTS & CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (SAA-C03)
- AWS Certified Developer – Associate (DVA-C02)
- AWS Certified Cloud Practitioner (CLF-C02)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Best Submission – TinyFish \$2M Pre-Accelerator Hackathon 2026
- Finalist – Solvit Hackathon 2025
- Top 100 / 7000+ Teams – Canara CyberSuraksha Hackathon
- Solved 500+ LeetCode problems.